

```
pyplot.legend()
pyplot.show()
```

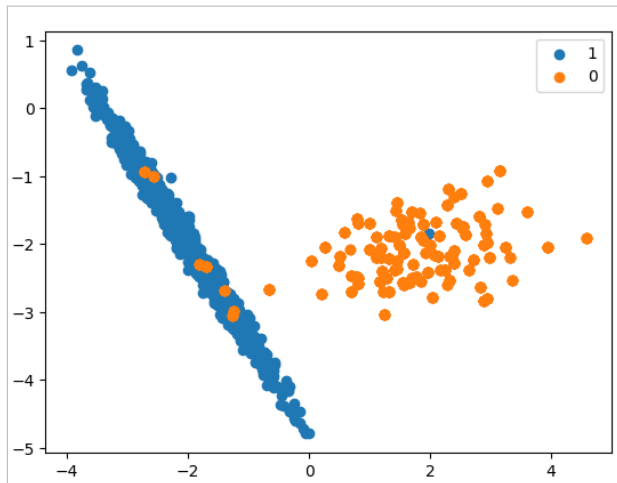


Figure 2: Scatter plot of examples by class label post using resampled data

RandomOverSampler

- Here we randomly select the rows from the minority class and perform duplication.
- *Logic*
 - Random oversampling involves duplicating examples from the minority class and adding them to the training dataset.
 - This is done by making exact copies of the minority class examples, which are selected randomly with replacement from the training dataset.
- The source code is:

```
from collections import Counter
from sklearn.datasets import make_classification
from imblearn.over_sampling import RandomOverSampler
from matplotlib import pyplot
from numpy import where

X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)
```

```
# summarise class distribution
print('Original dataset shape %s' % Counter(y))
Original dataset shape Counter({1: 894, 0: 106})
```

Now let us see how upsampling is performed:

```
ros = RandomOverSampler(random_state=24)
X_Res, y_Res = ros.fit_resample(X, y)
print('Resampled dataset shape %s' % Counter(y_Res))
Resampled dataset shape Counter({1: 894, 0: 894})
```

```
# scatter plot of examples by class label
for label, _ in Counter(y_Res).items():
    row_ix = where(y_Res == label)[0]
    pyplot.scatter(X_Res[row_ix, 0], X_Res[row_ix, 1],
label=str(label))
pyplot.legend()
pyplot.show()
```

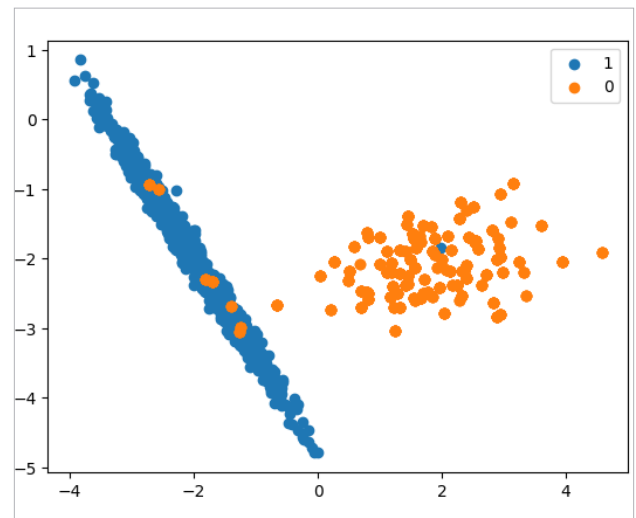


Figure 3: Scatter plot of examples by class label upsampled post using RandomOverSampler

SMOTE (Synthetic Minority Oversampling Technique)

- SMOTE is a widely used method for creating new examples. It involves selecting examples that are similar in the feature space, drawing a line between them, and generating a new sample at a point along that line.
- *Logic*
 - It works based on the K Nearest Neighbours algorithm, by generating the synthetic data points that fall under the existing minority class.
 - To be more specific, SMOTE randomly selects an instance from the minority class and identifies its k nearest minority class neighbours (usually k=5). Then, it randomly selects a neighbour from the k nearest neighbours, and creates a synthetic example at a randomly chosen point between the two examples in the feature space.
 - To generate the synthetic instances, SMOTE uses a convex combination of the two selected instances, a and b, which are connected by a line segment in the feature space.
 - Input records should not contain null values.
- Visualisation is shown in Figure 4.